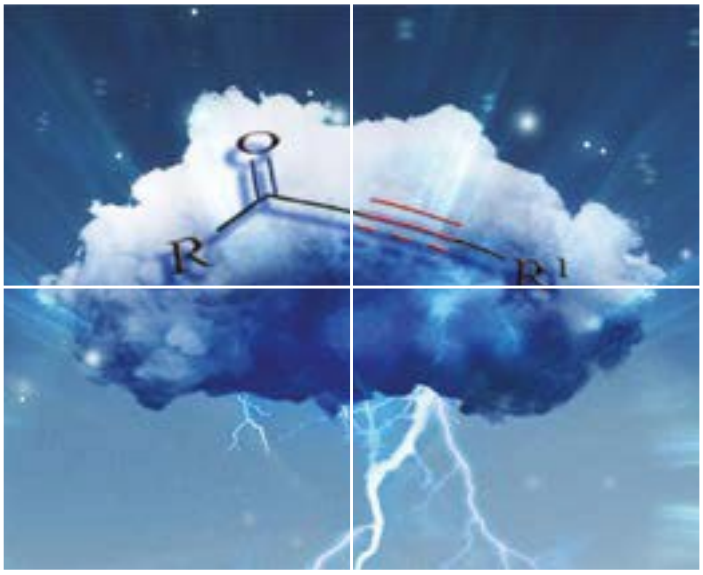
Check for updates

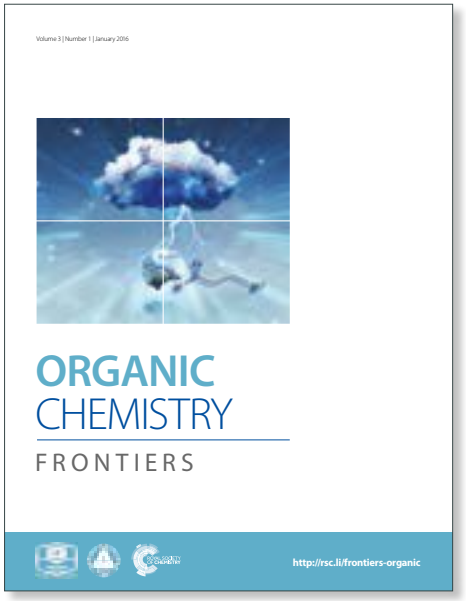
View Article Online
View Journal

ORGANIC CHEMISTRY FRONTIERS



Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: P. Wu, M. Jia and S. Ma, *Org. Chem. Front.*, 2019, DOI: 10.1039/C9QO00192A.



This is an Accepted Manuscript, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this Accepted Manuscript with the edited and formatted Advance Article as soon as it is available.

You can find more information about Accepted Manuscripts in the author guidelines.

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard Terms & Conditions and the ethical guidelines, outlined in our author and reviewer resource centre, still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this Accepted Manuscript or any consequences arising from the use of any information it contains.

Please do not adjust margins

View Article Online
DOI: 10.1039/C9QO00192A

Journal Name

ARTICLE

Pd-catalyzed coupling reaction of cyclobutanols with propargylic carbonates

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

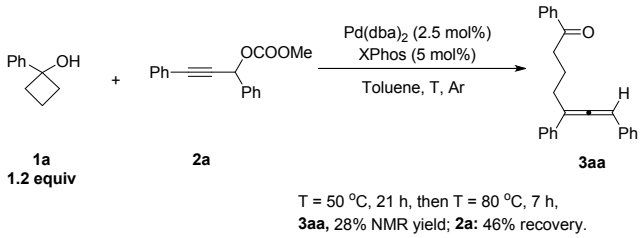
Penglin Wu,[a] Minqiang Jia*[a] and Shengming Ma*[a,b]

An efficient approach of Pd-catalyzed ring opening coupling reaction of cyclobutanols with propargylic carbonates was realized to provide a series of multisubstituted δ-allenyl ketones. The reaction had a wide substrate scope with different functional groups tolerated.

Introduction

Allene unit is an important type of molecular structure and found widespread in numerous natural products and bioactive molecules.[1] Thus, the synthesis of differently substituted allenes has attracted a lot of attention.[2] The coupling reaction of propargylic carbonates with different metal regents was an efficient method for the construction of multi-substituted allenes.[3] Recently, we found that cyclopropanols may be used as an efficient coupling reagent with propargylic carbonates for the synthesis of different substituted allenes. The reaction has a wide substrate scope under mild reaction conditions.[4] To expand the scope of such a reaction, we envisioned that cyclobutanols which are easily available from cyclobutanones,[5] could also be used in this type of coupling reaction via Pd catalysts as pioneered by Uemura.[6,7] In this paper, we report here a general method for the synthesis of different substituted δ-allenyl ketones from propargylic carbonates.

Firstly, standard reaction conditions with cyclopropanols were used for the coupling reaction of cyclobutanol **1a** with propargylic carbonate **2a**. Unfortunately, the reaction was stirred at 50 °C for 21 h and then 80 °C for 7 h to afford the desired allene product **3aa** in only 28% NMR yield with 46% of carbonate **2a** being recovered (Scheme 1). These results revealed that the reactivity of cyclobutanol was much lower than the cyclopropanol, which possibly due to their different ring strain (the ring strain of cyclopropane and cyclobutane are 29.0 and 26.3 kcal/mol, respectively[8]).

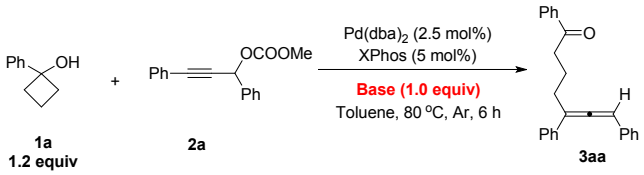


Scheme 1. Reaction of cyclobutanol **1a** with propargylic carbonate **2a** under standard conditions of cyclopropanol

Results and discussion

Based on these results, we decided to further screen the reaction conditions with cyclobutanol **1a** and propargylic carbonate **2a** as the model substrates. Firstly, different carbonate salts were added in toluene at 80 °C with $Pd(dba)_2$ and XPhos as the catalyst. K_2CO_3 and Cs_2CO_3 provided the highest yields (Table 1, entries 1-4). $K_3PO_4 \cdot 3H_2O$ had almost the same effect (Table 1, entry 5). Other bases such as *t*-BuOK, $NaHCO_3$, CsOPiv, DIPEA, and Et_3N were also tested demonstrating inferior results (Table 1, entries 6-10). Finally, we also found that the very freshly prepared carbonate **2a** can be used directly to give the corresponding allene **3aa** in 80% NMR yield without addition of any base (Table 1, entry 11). However, 1.0 equiv of base was required for the (slightly) aged **2a**.

Table 1. The effect of base[a]



[a] Department of Chemistry, Fudan University, 220 Handan Road, Shanghai 200433, P. R. China
[b] State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China. E-mail: masm@sioc.ac.cn
† Electronic Supplementary Information (ESI) available: See
DOI: 10.1039/x0xx00000x

Published on 01 April 2019. Downloaded by State University of New York at Stony Brook on 4/1/2019 3:31:56 PM.

Organic Chemistry Frontiers Accepted Manuscript

Please do not adjust margins

View Article Online
DOI: 10.1039/C9QO01048A

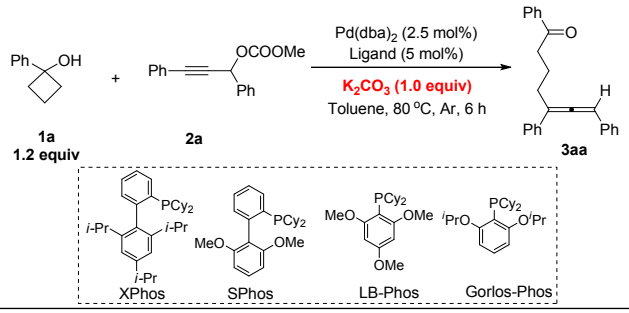
Published on 01 April 2019. Downloaded by State University of New York at Stony Brook on 4/1/2019 3:31:56 PM.

entry	Base	NMR yield of **3aa** (%)[b]	Recovery of **2a** (%)[b]
1	Li_2CO_3	59	3
2	Na_2CO_3	54	10
3	K_2CO_3	80	0
4	Cs_2CO_3	80	0
5	$K_3PO_4 \cdot 3H_2O$	79	0
6	t-BuOK	complicated	-
7	$NaHCO_3$	55	12
8	CsOPiv	0	96
9	DIPEA	51	12
10	Et_3N	61	3
11[c]	--	80	0

[a] Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), $Pd(dba)_2$ (2.5 mol%), XPhos (5.0 mol%), base (1.0 equiv) in toluene (1.0 mL) at 80 °C. [b] NMR yield with CH_2Br_2 as the internal standard. [c] The carbonate **2a** was used immediately after chromatography on silica gel, reaction time: 3 h.

Then, a range of different phosphine ligands were screened. The use of PPh_3 and dppp led to complicated results (Table 2, entries 1-2). When other phosphine ligands such as PCy_3, dppf, dppm, LB-Phos and Gorlos-Phos were used, the starting material **2a** was recovered in 36-75% with no product **3aa** being observed (Table 2, entries 3-7). Fortunately, the SPhos and *rac*-BINAP resulted in a moderate yield of the product **3aa** with 5-10% yield of **2a** remaining (Table 2, entries 8-9). Finally, it was observed that XPhos was still the best ligand, which provided the allene product in 80% yield and no **2a** was recovered (Table 2, entry 10).

Table 2. The effect of ligand[a]

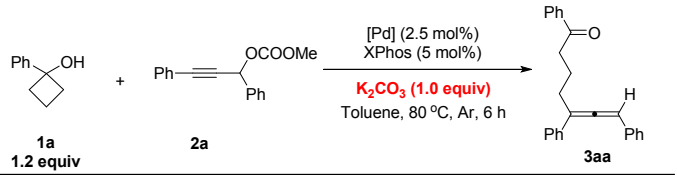


entry	Ligand	NMR yield of **3aa** (%)[b]	Recovery of **2a** (%)[b]
1	PPh_3	Complicated	-
2	dppp	Complicated	-
3	PCy_3	0	75
4	dppf	0	74
5	dppm	0	36
6	LB-Phos	0	63
7	Gorlos-Phos	0	39
8	SPhos	56	10
9	*rac*-BINAP	63	5
10	**XPhos**	**80**	**0**

[a] Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), $Pd(dba)_2$ (2.5 mol%), Ligand (5.0 mol%), K_2CO_3 (1.0 equiv) in toluene (1.0 mL) at 80 °C. [b] NMR yield with CH_2Br_2 as the internal standard.

Subsequently, with XPhos as the optimal ligand, Pd(II) salts were tested. Different Pd(II) sources such as $Pd(OAc)_2$, $PdCl_2$ and $Pd(TFA)_2$ gave 9-28% yields of the product **3aa** with 31-81% recovery of the starting material **2a** (Table 3, entries 2-4).

Table 3. The effect of Pd-catalyst[a]

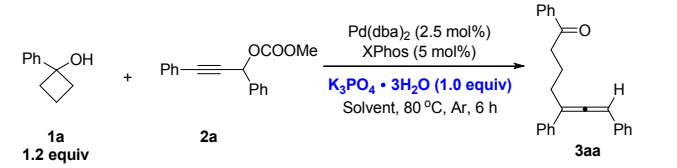


entry	[Pd]	NMR yield of **3aa** (%)[b]	Recovery of **2a** (%)[b]
1	**$Pd(dba)_2$**	**80**	**0**
2	$Pd(OAc)_2$	28	31
3	$PdCl_2$	13	73
4	$Pd(TFA)_2$	9	81

[a] Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), [Pd] (2.5 mol%), Xphos (5.0 mol%), K_2CO_3 (1.0 equiv) in toluene (1.0 mL) at 80 °C. [b] NMR yield with CH_2Br_2 as the internal standard.

Considering of the convenience, $K_3PO_4 \cdot 3H_2O$ was used as the base for further study. Solvents were further screened. THF, dioxane, CH_3CN and DCM all gave the desired product **3aa** albeit with lower yields (49-69%, Table 4, entries 2-5). When DMF was used, the reaction was complicated and 14% of **3aa** was observed (Table 4, entry 6).

Table 4. The effect of solvent[a]



entry	Solvent	NMR yield of **3aa** (%)[b]	Recovery of **2a** (%)[b]
1	**toluene**	**79**	**0**
2	THF	69	0
3	dioxane	61	0
4	CH_3CN	56	0
5	DCM	49	0
6	DMF	14	-

[a] Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), $Pd(dba)_2$ (2.5 mol%), Xphos (5.0 mol%), K_2CO_3 (1.0 equiv) in solvent (1.0 mL) at 80 °C. [b] NMR yield with CH_2Br_2 as the internal standard.

Finally, lowering the temperature to 70 °C or 60 °C can both provide 81% yield of **3aa** (Table 5, entries 2-3). However, the reaction at 60 °C on 1 mmol scale reaction, led to the minor recovery of the propargylic carbonate **2a**. Decreasing the temperature further to 50 °C resulted in a lower yield of **3aa** with 53% recovery of starting material **2a** (Table 5, entry 4).

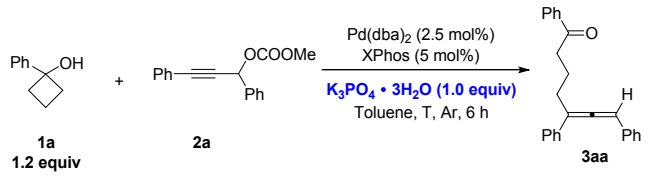
Organic Chemistry Frontiers Accepted Manuscript

This journal is © The Royal Society of Chemistry 20xx

Please do not adjust margins

View Article Online DOI: 10.1039/C9QO00192A

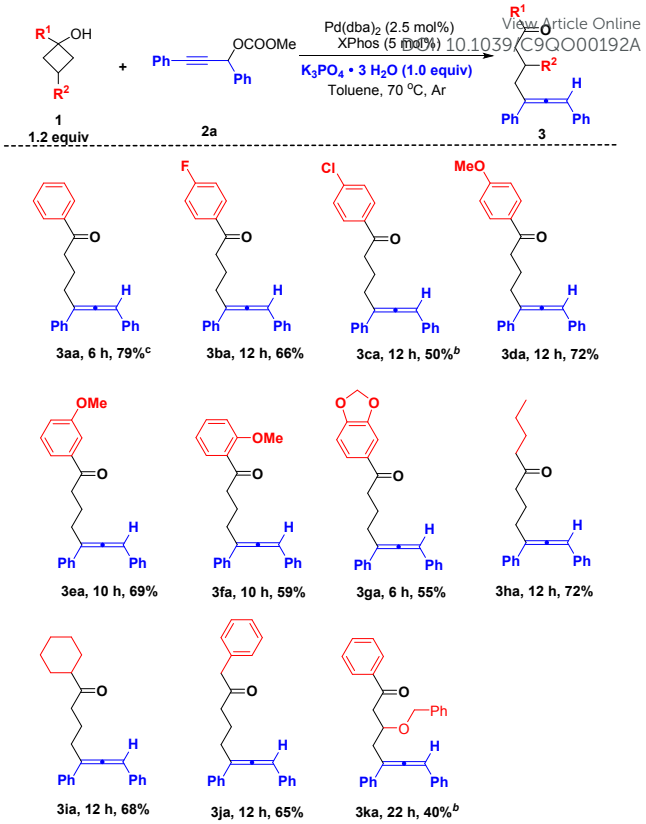
Table 5. The effect of temperature[a]



entry	Temperture	NMR yield of **3aa** (%)[b]	Recovery of **2a** (%)[b]
1	80	79	0
2	**70**	**81**	**0**
3	60	81	0
4	50	31	53

[a] Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), $Pd(dba)_2$ (2.5 mol%), Xphos (5.0 mol%), K_2CO_3 (1.0 equiv) in Toluene (1.0 mL). [b] NMR yield with CH_2Br_2 as the internal standard.

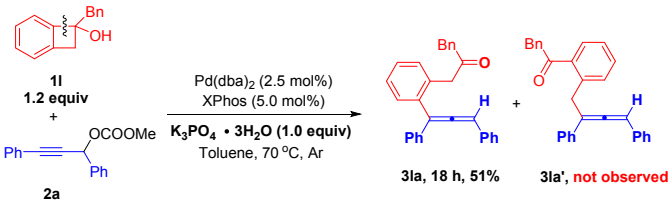
With the optimized reaction conditions above (Table 5, entry 2), substrate scope of various cyclobutanols **1** with the propargylic carbonates **2a** was investigated. All the reactions were carried out on 1 mmol scale and results are summarized in Scheme 2. When the R^1 is aryl group, besides phenyl, the substituent group on the phenyl group can be either electron-withdrawing group (F, Cl) or electron-donating group (OMe), providing the allene products in 50-79% yields (Scheme 2, **3aa-3ga**). In addition, the substituents on the phenyl rings at the *ortho, meta, and para* positions may all be tolerated (Scheme 2, **3da-3fa**). And the piperonyl group can be tolerated well providing 55% yield of **3ga**. Except for the aryl group, R^1 can be different alkyl groups such as *n*-butyl, cyclohexyl and benzyl to provide the desired allene product in 65-72% yields (Scheme 2, **3ha-3ja**). The reaction also woked with a cyclobutanol with an additional substituent group at 3-position provding the expected product **3ka** with 40% yield (Scheme 2).



[a] Reaction conditions: **1** (1.2 mmol), **2a** (1.0 mmol), $Pd(dba)_2$ (2.5 mol%), XPhos (5.0 mol%), $K_3PO_4 \cdot 3H_2O$ (1.0 equiv) in toluene (5.0 mL) at 70 °C. [b] $Pd(dba)_2$ (5.0 mol%) and XPhos (10.0 mol%) was used. [c] Byproduct, 1,3-dipenylpropdiene, was formed in 4% NMR yield.

Scheme 2. Substrate scope referred to cyclobutanols[a]

Notably, when benzocyclobutanol **1l** was used in the reaction, the product **3la** with the cleavage of the proximal bond was selectively generated in 51% yield, and the distal bond cleavage product **3la'** was not observed (Scheme 3).



[a] Reaction conditions: **1l** (1.2 mmol), **2a** (1.0 mmol), $Pd(dba)_2$ (2.5 mol%), XPhos (5.0 mol%), $K_3PO_4 \cdot 3H_2O$ (1.0 equiv) in toluene (5.0 mL) at 70 °C.

Scheme 3. The reaction of benzocyclobutanol[a]

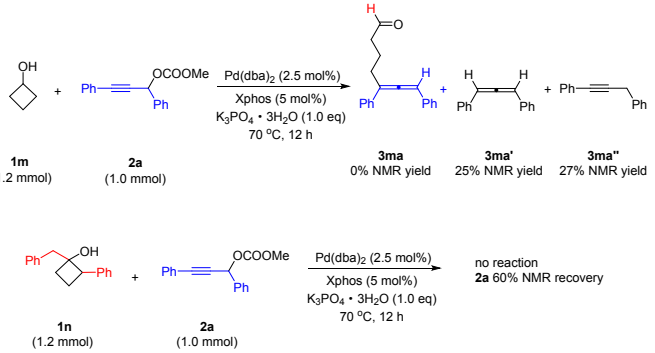
However, when R^1=H, the reaction did not provide the expected product **3ma,** instead, by-product 1,3-diphenypropdiene **3ma'** and 1,3-dipenylpropyne **3ma"** were formed in 25% and 27% NMR yields, respectively.[4] The reaction of the cyclobutanol **1n,** which has an additional substituent at the 2-position, gave no allene product with 60% recovery of the starting material **2a** (Scheme 4).

Published on 01 April 2019. Downloaded by State University of New York at Stony Brook on 4/1/2019 3:31:56 PM.

Organic Chemistry Frontiers Accepted Manuscript

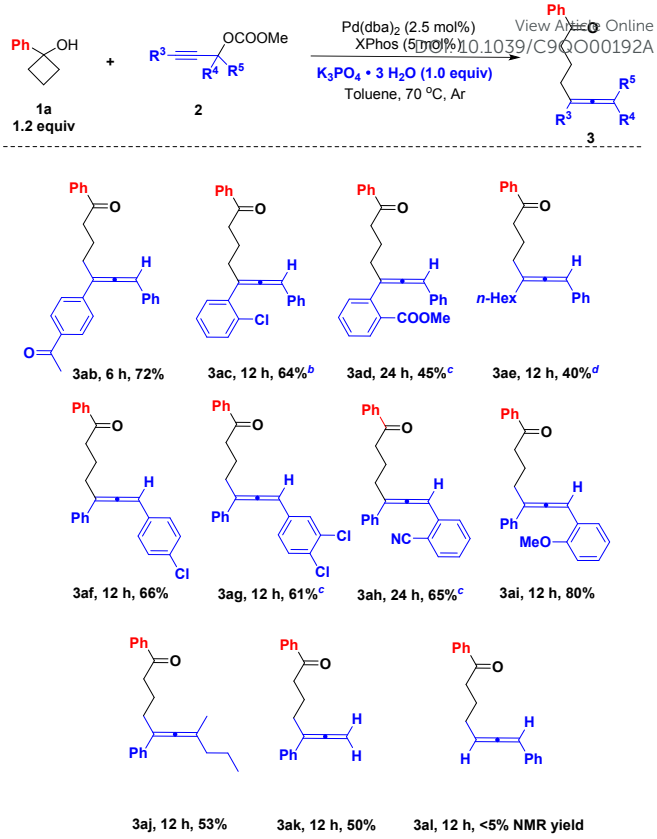
Please do not adjust margins

View Article Online
DOI: 10.1039/C9QO00192A



Scheme 4. Failed substrates of cyclobutanols

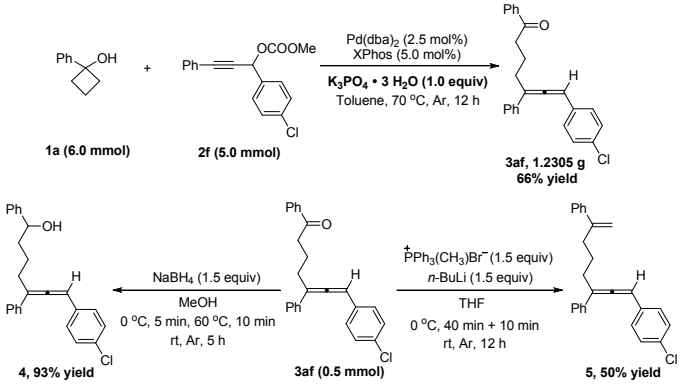
Next, the scope of propargylic carbonates **2** was also tested. When the R^5 is H, the R^3, R^4 substituents can be different aryl or alkyl groups. For example, 4-acetyl, 2-chloro, and 2-methoxycarbonyl substituted phenyls at the R^3 position were tolerated providing the allene products **3ab-3ad** in 45-72% yields. R^3 could also be *n*-hexyl resulted in 40% yield of **3ae** by decreasing the amount of **1a** to 1.0 mmol and increasing the amount of propargylic carbonate **2e** to 1.5 mmol. And 4-chloro, 3,4-dichloro, 2-cyano and 2-methoxy substituted phenyls could be introduced successfully to the R^4 position to give the final product **3af-3ai** in 61-80% yields. It should be noted that the catalyst loading should be improved to 5 mol% for **3ac**, **3ad**, **3ag** and **3ah** to make sure the completely consumption of the starting carbonates. Notably, besides the secondary propargylic carbonates, tertiary propargylic carbonates (R^2 = Ph, R^3 = ^{n}Pr, R^4 = Me) could also be used to afford the fully substituted allene products **3aj** in moderate yield. Besides, when R^4 = R^5= H, the reaction afforded **3ak** with 50% yield. However, When R^3 = H, the corresponding allene product **3al** was formed in a very low yield (<5% by ^{1}H NMR) (Scheme 5).



[a] Reaction conditions: **1a** (1.2 mmol), **2** (1.0 mmol), $Pd(dba)_2$ (2.5 mol%), XPhos (5.0 mol%), $K_3PO_4 \cdot 3H_2O$ (1.0 equiv) in toluene (5.0 mL) at 70 °C. [b] **1a** (1.0 mmol), **2** (1.2 mmol), $Pd(dba)_2$ (5.0 mol%) and XPhos (10.0 mol %) was used. [c] $Pd(dba)_2$ (5.0 mol%) and XPhos (10.0 mol%) was used. [d] **1a** (1.0 mmol) and **2** (1.5 mmol) was used.

Scheme 5. Substrate scope referred to propargylic carbonates[a]

In order to demonstrate the synthetic utility and practicality of the method, the allene product **3ab** was synthesized on gram scale. The 5.0 mmol scale reaction was carried out providing 1.2305 g of the desried product **3ab** in 66% isolated yield. And the allene product **3ab** can be subjected to some transformations. For example, it can be reduced to the secondary alcohol **4** with $NaBH_4$, and react with the Wittig reagent producing the product **5** (Scheme 6).



Scheme 6. Gram-scale reaction and synthetic utilities.

Organic Chemistry Frontiers Accepted Manuscript

Published on 01 April 2019. Downloaded by State University of New York at Stony Brook on 4/1/2019 3:31:56 PM.

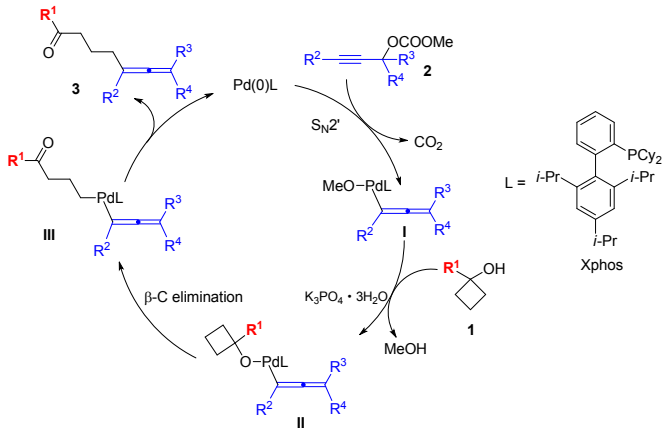
This journal is © The Royal Society of Chemistry 20xx

Please do not adjust margins

View Article Online
DOI: 10.1039/C9QO00192A

Published on 01 April 2019. Downloaded by State University of New York at Stony Brook on 4/1/2019 3:31:56 PM.

Based on the previous experimental results and knowledge,[4,9] a possible mechanism was proposed as shown in Scheme 7. Firstly, the propargylic carbonate **2** reacted with Pd(0) *via* SN2'-type oxidative addition together with the elimination of a molecule of CO_2, which leads to the allenyl palladium methoxide intermediate **I**. Then the intermediate **I** underwent ligand exchange with cyclobutanol **1** generating the intermediate **II**, which would provide intermediate **III** by β-C elimination. The intermediate **III** delivered the final product **3** *via* reductive elimination and regenerated the Pd(0) species completing the catalytic cycle.



Scheme 7. The proposed mechanism.

Conclusions

In summary, an efficient method to synthesize the different multi-substituted δ-allenyl ketones via palladium-catalyzed ring opening of cyclobutanols and oxidative addition of propargylic carbonates was developed. The reaction had a wide substrate scope with different functional groups tolerated, and the method may be easily expanded to gram scale synthesis with potential synthetic utility.

Acknowledgements

Financial support from the National Basic Research Program of China (2015CB856600) and the National Natural Science Foundation of China (21502020) is greatly appreciated. We thank Yifan Cui of our group for reproducing the results of **3ca**, **3ga**, and **3ab** represented in this study.

Notes and references

1 For a review on allenes in natural products and bioactive molecules, see: A. Hoffmann-Röder and N. Krause, *Angew. Chem., Int. Ed.* 2004, **43**, 1196.

2 For reviews on the synthesis of different substituted allenes, see: (a) A. Hoffmann-Röder and N. Krause, *Angew. Chem., Int. Ed.* 2002, **41**, 2933; (b) N. Krause, A. Hoffmann-Röder, *Tetrahedron* 2004, **60**, 11671; (c) G. B. Hammond, *ACS Symp. Ser.* 2005, **911**, 204; (d) K. M. Brummond, J. E. DeForrest, *Synthesis* 2007, 795; (e) M. Ogasawara, Tetrahedron: *Asymmetry* 2009, **20**, 259; (f) S. Yu, S. Ma, *Chem. Commun.* 2011, **47**, 5384; (g) R. K. Neff, D. E. Frantz, *ACS Catal.* 2014, **4**, 519; (h) J. Ye, S. Ma, *Org. Chem. Front.* 2014, **1**, 1210; (i) R. K. Neff, D. E. Frantz, *Tetrahedron* 2015, **71**, 7.

3 Selected examples, see: (a) E. Keinan, E. Bosch, *J. Org. Chem.* 1986, **51**, 4006; (b) P. Dixneuf, T. Guyot, M. Ness, S. Roberts, *Chem. Commun.* 1997, 2083; (c) R. Riveiros, D. Rodriguez, J. Sestelo, L. Sarandeses, *Org. Lett.* 2006, **8**, 1403; (d) H. Ito, Y. Sasaki, M. Sawamura, *J. Am. Chem. Soc.* **2008**, 130, 15774; (e) H. Ohmiya, H. Ito, M. Sawamura, *Org. Lett.* 2009, **11**, 5618; (f) Q. Li, Jeng, J. Jeng, H. Gau, Eur. *J. Org. Chem*. 2014, 7916; (g) T. Zhao, Y. Yang, T. Lessing, K. Szabó, *J. Am. Chem. Soc.* 2014, **136**, 7563. (h) H. Luo, Y. Yu, S. Ma, *Org. Chem. Front.* 2016, **3**, 1705. (i) S. Kessler, J. Bäckvall, *Angew. Chem. Int. Ed.* 2016, **55**, 3734; (j) Q. Lu, S. Greßies, F. J. R. Klauck, F. Glorius, *Angew. Chem., Int. Ed.* 2017, **56**, 6660.

4 P. Wu, M. Jia, W. Lin, S. Ma, *Org. Lett*. 2018, **20**, 554.

5 (a) K. Jia, F. Zhang, H. Huang, Y. Chen, *J. Am. Chem. Soc.* 2016, **138**, 1514; (b) B. Casey, C. Eakin, R. Flowers II, *Tetrahedron Lett*. 2009, **50**, 1264; (c) H. Zeng, P. Pan, J. Chen, H. Gong, C. Li, *Eur. J. Org. Chem.* 2017, 1070.

6 For examples on coupling reaction of the ring-open of cyclubutanols via Pd catalysis, see: (a) T. Nishimura, K. Ohe, S. Uemura, *J. Am. Chem. Soc.* 1999, **121**, 2645; (b) T. Nishimura, S. Uemura, *J. Am. Chem. Soc*. 1999, **121**, 11010; (c) T. Nishimura, K. Ohe, S. Uemura, *J. Org. Chem*. 2001, **66**, 1455; (d) T. Nishimura, S. Matsumura, Y. Maeda, S. Uemura, *Chem. Commun.* 2002, 50.; (e) T. Nishimura, S. Matsumura, Y. Maeda, S. Uemura, *Tetrahedron Lett*. 2002, **43,** 3037; (f) M. Ethirajan, H. Oh, J. Cha, *Org. Lett*. 2007, **9,** 2693; (g) A. Chtchemelinine, D. Rosa, A. Orellana, *J. Org. Chem.* 2011, **76**, 9157; (h) A. Ziadi, R. Martin, *Org. Lett.* 2012, **14**, 1266; (i)A. Ziadi, A. Correa, R. Martin, *Chem. Commun.* 2013, **49**, 4286; (j) L. Chen, F. Sun, Y. Sun, Z. Xu, Z. Zheng, Y. Cui, J. Cao, L. Xu, *Adv. Synth. Catal.* 2018, **360**, 411.

7 S. Matsumura, Y. Maeda, T. Nishimura, S. Uemura, *J. Am. Chem. Soc.* 2003, **125**, 8862.

8 P. Khoury, W. Tam, J. Goddard, *Tetrahrdron* 2004, **60,** 8103.

9 For a review, see: (a) J. Tsuji, T. Mandai, *Angew. Chem., Int. Ed. Engl.* 1995, **34**, 2589; See also: (b) S. Ogoshi, K. Tsutsumi, M. Ooi, H. Kurosawa, *J. Am. Chem. Soc.* 1995, **117**, 10415; (c) T. Konno, M. Tanikawa, T. Ishihara, H. Yamanaka, *Chem. Lett.* 2000, 1360; (d) B. M. Trost, L. Debien, *Chem. Sci.* 2016, **7**, 4985.

Organic Chemistry Frontiers Accepted Manuscript

Please do not adjust margins